

FREE INTRODUCTORY EDITION

AI-Level

The Five Circles of Our Future

From Today's AI to a Type II Civilization

How can artificial intelligence help humanity solve harder problems, build greater capabilities, and expand its possibilities?

This short book introduces five circles of progress: software, computing hardware, physical industry, biology, and stellar infrastructure. Each circle contains grand challenges worth a lifetime of work.

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A better future begins with a problem worth solving.

From answers to capabilities

Imagine asking an AI to design a machine that repairs a water pump. A convincing answer is a beginning. Someone must still make the parts, assemble the machine, test it under difficult conditions, and keep it working. If the result succeeds, a community gains something lasting: the ability to restore its water supply.

That journey is the central idea of this book. Intelligence matters when it becomes a capability people can use.

What does AI-Level mean?

AI-Level asks what a system can actually accomplish, how it learns, and how reliably it improves. Reasoning, memory, learning, and organization each matter. A system may be strong in one and weak in another. One score cannot tell the whole story.

Remembering a correction differs from learning a method that prevents future mistakes. Producing a discovery differs from confirming it and using it successfully. An organization improves when useful knowledge becomes part of its ordinary work, even after the original contributors move on.

The same practical cycle appears repeatedly: **identify a problem, implement a solution, validate the result, and put it to use.** What happens next supplies evidence for the following attempt. Humans and AI can share these tasks in many ways; the degree of autonomy is a separate question.

Why five circles?

A circle describes the domain in which intelligence acts. Software changes instructions. Hardware changes the machines that compute. Physical industry changes how we make and maintain things. Biology changes our ability to understand and help living systems. Stellar infrastructure extends productive activity beyond Earth.

These circles form a proposed framework for organizing research and action. They overlap and develop together. Their numbers do not rank human worth, and progress through them is not guaranteed.

For each circle, ask three questions: **What can we build? What still blocks us? What evidence would show that we have moved forward?**

Circle I: Software

Turn useful answers into reliable work

A large language model, or LLM, generates language by learning patterns from data. Combined with tools, memory, and a process for taking action, it can become part of an AI agent: a system that carries out steps toward a goal.

The challenge is making that work dependable. A research assistant should track evidence, notice mistakes, and improve its methods. A coding agent should leave behind software that works under real use. The circle's ambition is sustained improvement that survives careful testing.

Eight grand problems

- **Memory and lasting learning.** Retain useful experience and turn it into better performance on new tasks without losing earlier abilities.
- **Reliable reasoning and action.** Complete long tasks, recover from errors, and recognize when the system lacks enough information to continue.
- **Verified self-improvement.** Change software or reasoning methods and demonstrate that the changes improve performance beyond the examples used to develop them.
- **Better ways to improve.** Improve how the system searches for, tests, and selects changes to its own methods.
- **Discovery that becomes capability.** Produce a new finding, confirm it, and show that incorporating it improves later work.
- **Trustworthy evaluation.** Build tests that detect hidden failures and resist shortcuts, including tests developed independently of the proposed solution.
- **Organizations that learn.** Help people and agents coordinate, challenge one another's mistakes, and turn successful methods into lasting practices.
- **Honest measurement.** Compare useful results alongside time, cost, resources, and human assistance so that claims remain meaningful.

What progress would look like

A software system repeatedly improves a useful tool. Independent assessment confirms gains on unfamiliar tasks, checks that earlier capabilities remain intact, and records the resources and human help required. The improved tool is then used successfully.

Why it matters now: More dependable software can support education, research, public services, and almost every other circle.

Circle II: Computing hardware

Build the machines that make intelligence possible

Every AI system runs on physical equipment. A better chip design becomes useful only when it can be manufactured, supplied with power, cooled, maintained, and connected to a working system.

This circle asks whether intelligence can help improve that entire chain. The important outcome is usable computing capacity at an acceptable cost. A faster design that cannot be manufactured reliably may contribute less than a modest improvement that works across thousands of machines.

Eight grand problems

- **Useful chip design.** Design processors and systems that meet real needs while respecting manufacturing, energy, and cost constraints.
- **Reliable fabrication.** Control the many production steps needed to turn a design into working devices.
- **Measurement and yield.** Find defects, explain changes in production, and increase the share of manufactured chips that work correctly.
- **Materials and connections.** Develop devices, materials, packaging, and links that move and process information effectively.
- **More computation per unit of energy.** Reduce the energy needed for useful work across chips, memory, and whole systems.
- **Cooling and dependable operation.** Remove heat, manage failures, and keep equipment useful over its working life.
- **Equipment and essential inputs.** Secure and improve the tools, materials, maintenance, and specialist components on which production depends.
- **Complete hardware integration.** Bring newly produced hardware into working computing systems and verify the benefits in actual operation.

What progress would look like

A team designs a component, manufactures it with acceptable yield, integrates it into a working system, and demonstrates a useful improvement after counting energy, equipment, and maintenance costs. It identifies which capabilities still depend on outside suppliers.

Why it matters now: Efficient, dependable computing can make useful AI and scientific tools more accessible. Circle II refers to computing hardware; the separate term "Type II civilization" refers to stellar-scale energy use.

Circle III: Physical industry

Make, maintain, and expand productive capacity

A robot that performs one carefully arranged task has demonstrated something valuable. A productive industrial system must do much more: handle imperfect materials, diagnose failures, replace worn parts, and keep producing useful outputs.

The ambition here is a growing ability to build and maintain the tools civilization relies on. Consider a remote workshop. Its capability expands when it can repair equipment, manufacture more of its own replacement parts, and build additional working machines from declared supplies.

Eight grand problems

- **Robots in varied environments.** Perceive, manipulate, and move safely and reliably when objects and conditions differ from rehearsed examples.
- **Diagnosis and repair.** Identify why equipment fails, perform the repair, and verify that normal operation has returned.
- **Raw materials to useful products.** Turn available resources into consistent materials, parts, and finished goods with manageable waste.
- **Precision and complex machinery.** Manufacture and assemble accurate components, instruments, and machines that simpler tools cannot readily replace.
- **Coordinated supply and logistics.** Get the right materials, parts, and resources to the right place despite interruptions and changing demand.
- **Reproducing productive capacity.** Build additional functioning production equipment while openly accounting for imported components and outside assistance.
- **Dependable energy infrastructure.** Build, operate, and maintain the power supply and supporting systems that industry needs.
- **Physical verification.** Measure the quality, durability, and safety of what is produced under conditions that reflect actual use.

What progress would look like

A workshop produces useful goods, maintains essential equipment, and builds a functioning successor machine. Repeated operation establishes its quality and output. The record states which electronics, tools, materials, energy, and human skills still come from elsewhere.

Why it matters now: Better industrial capability can help communities maintain water systems, housing, transport, and energy infrastructure. Those benefits matter wherever the longer journey leads.

Circle IV: Biology

Turn knowledge of life into healthier lives

Living systems are complex. A pattern in data can suggest a promising explanation, but a useful intervention requires evidence that changing something produces the intended benefit. Results must also hold across relevant conditions and reach the people who need them.

AI can contribute hypotheses, experimental designs, and tools for analysis. This circle's mission is to connect those contributions to reproducible science and meaningful outcomes. Biology advances alongside the other circles, with health and well-being as purposes in their own right.

Eight grand problems

- **Causal understanding.** Discover which biological mechanisms drive an outcome and which observations merely accompany it.
- **Reliable experimental laboratories.** Connect planning, experiments, measurement, and analysis while making important findings reproducible by independent teams.
- **Cancer.** Improve understanding, detection, and treatment across diverse cancers, including resistance, recurrence, and differences between patients.
- **Aging and healthy life.** Understand processes associated with aging and test ways to preserve function and extend healthy years.
- **Repair and regeneration.** Restore damaged tissues and organs while controlling unwanted growth and other adverse effects.
- **Brain and nervous system disorders.** Develop better explanations, measurements, and interventions for complex neurological and degenerative conditions.
- **Interventions that reach their targets.** Deliver treatments where they are needed with suitable durability, precision, and tolerability.
- **Benefits at population scale.** Make validated interventions practical and accessible, and learn from monitoring their long-term outcomes.

What progress would look like

A clearly defined intervention produces a meaningful benefit that independent work confirms. Evaluation follows it from relevant experiments through appropriate human studies and delivery, with continued attention to durability and adverse effects. The evidence supports a specific claim about who benefits and under what conditions.

Why it matters now: Reducing suffering and preserving the ability to live well are direct goals. The problems above are research ambitions; listing them does not imply that a cure or a completion date is known.

Circle V+: Stellar infrastructure

Build a durable future beyond Earth

A Type II civilization, in the Kardashev classification, uses energy on the scale of its star. One proposed route is a Dyson swarm: many separate structures orbiting a star and collecting some of its energy. The familiar image of a rigid shell is not the engineering approach used here.

Such a future would require much more than collectors. It would need supplies, manufacturing, maintenance, useful energy demand, and coordination across enormous distances. An early space industry could advance these capabilities while remaining far below Type II scale.

Eight grand problems

- **Supplies beyond Earth.** Obtain and process useful materials in space while managing transport costs and remaining dependence on Earth.
- **Orbital construction and growth.** Manufacture and assemble infrastructure that adds more useful capacity than is lost through wear and failure.
- **Durable energy collectors.** Build structures that collect stellar energy and remain useful under harsh operating conditions.
- **Useful energy conversion and delivery.** Convert collected energy into forms that can power industry, computation, transport, or other valuable activity.
- **Waste heat.** Design systems that can release heat effectively; collecting more energy also creates a larger thermal engineering problem.
- **Stable orbits and collision avoidance.** Coordinate motion, manage debris, and prevent local failures from damaging a growing network.
- **Maintenance over long periods.** Inspect, repair, replace, and upgrade infrastructure when replacement supplies and human assistance may be distant.
- **Organization across distance.** Combine local decision-making with wider coordination when communication delays prevent continuous central control.

What progress would look like

An orbital production system expands useful operating capacity over a declared period after accounting for failures, maintenance, and imported supplies. That would be an important milestone toward a larger space economy. Reaching Type II would require a vastly greater energy scale.

Why the distinction matters: Energy use measures one capability of a civilization. It does not establish intelligence, wisdom, fairness, or the quality of anyone's life.

How we move between circles

Solving a hard problem changes what becomes possible next. More reliable software can support better instrument design. Better instruments can improve experiments. Better manufacturing can turn a successful prototype into a capability that many people can use.

This is the meaning of moving outward: increasing the range of problems we can solve and the scale at which we can sustain solutions.

Follow the constraint

Imagine a team building an automated laboratory. At first, unreliable planning wastes experiments. After the planning improves, inconsistent sample handling may become the main obstacle. Once that improves, the team may discover that it cannot measure the biological outcome accurately enough.

The center of the work has shifted. The team should examine what now limits its mission and direct effort there. In this book, **centering on a circle** means giving a domain sustained attention because its constraints now govern the next meaningful advance in a particular program.

That decision requires evidence. Ask: if this step became faster, cheaper, or more reliable, would the overall result improve? Sometimes a celebrated technology makes little difference because a less visible constraint still determines the outcome.

Work in parallel

Humanity does not have to finish every software problem before making a better chip. Biology does not need to wait for fully automated factories. Hardware and industry help develop one another. Several circles can deserve sustained effort at the same time.

"Circle time" can therefore mean a period of committed work on the problems that matter most to a mission. It is a way to organize attention, investment, and cooperation. It is not a fixed calendar dividing the future into predetermined eras.

Keep the purpose visible

A grand problem becomes manageable through smaller demonstrations. Each should name the capability sought, the evidence required, and the dependencies that remain. Success should open a useful next step.

People must also ask who gains access, who bears the costs, and whether the capability serves a worthwhile purpose. Greater technical reach creates more choices. Human judgment helps determine which choices deserve pursuit.

Choose a problem worth your time

You do not need to solve an entire circle to contribute. A dependable measurement, a repaired machine, a reproducible experiment, or a better teaching method can remove a constraint that holds back many other people.

Start with a need you understand. Look for a failure that recurs, an expensive step that prevents wider use, or a result that others cannot reproduce. Learn why earlier attempts struggled. Then define a contribution small enough to test and important enough to matter.

A problem card

Write five short answers before committing to a project:

1. **Purpose:** Whose situation improves if this problem is solved?
2. **Constraint:** What currently prevents that improvement, and what evidence points to it?
3. **Contribution:** What specific capability can you or your team build?
4. **Demonstration:** What observation would establish progress, and who can check it?
5. **Next use:** How can someone adopt the result and build on it?

For example, a team might help small workshops diagnose one common pump failure. Its first contribution could be a tested diagnostic method and repair procedure. A meaningful demonstration would use unfamiliar faulty pumps, record unsuccessful repairs, and establish whether other technicians can apply the method. The result would be modest in scale and useful in the world.

Many roles matter

Researchers can sharpen questions and test explanations. Engineers and technicians can make systems work outside demonstrations. Teachers can help more people understand and contribute. Entrepreneurs can turn validated capabilities into dependable services. Funders and public institutions can support shared equipment, careful evaluation, and work whose benefits take time to emerge.

Communities and users contribute knowledge about real needs and whether a proposed solution helps. Their experience belongs in the process from the beginning.

Choose a challenge. Find collaborators. State what success would look like. Report failures as well as gains. Make useful results understandable enough for others to evaluate and build upon.

The invitation is to give difficult, worthwhile problems sustained attention. A civilization advances through many contributions that become dependable enough for the next person to use.

Continue the journey

The five circles connect an ordinary question about AI to an extraordinary question about humanity: how far can we extend our ability to understand, build, repair, and care?

The answer will emerge through work. Some discoveries will open unexpected routes. Some ambitious projects will fail. Progress will depend on our willingness to test ideas, revise priorities, and make successful capabilities useful beyond the team that first creates them.

A Dyson swarm is a distant horizon. The work along the way includes reliable tools, efficient computing, resilient infrastructure, and better health. Those achievements already offer reasons to begin.

Inside the full book

AI-Level: From LLMs to a Type II Civilization

The Five Circles and the Grand Challenges on the Road to a Dyson Swarm

The planned full book develops this framework across 32 chapters, examining how systems and organizations learn, how to assess improvement, and how the 40 grand problems depend on one another. It explores engineering approaches, evidence for major milestones, and institutions that could sustain the work.

Explore the ideas and follow the project

Visit the [AI-Level project on GitHub](#) for the underlying framework and public book materials. Share the project with someone working on a problem in one of the circles. Bring questions, counterexamples, and clearer proposals: a useful framework should improve through criticism and evidence.

Further reading

- Chengshuai Yang, [Circle-Indexed Individual and Organizational Intelligence](#). The source framework behind this introduction.
- [Public book materials](#). Outlines and developing material for the AI-Level book project.
- Jason T. Wright, [Dyson Spheres](#). Scientific background on the stellar-scale structures discussed in Circle V+.

This introduction presents the author's research agenda in simplified language. Its grand problems are proposed priorities, and its milestones are examples of evidence to seek. They are not claims that the capabilities have already been achieved or predictions of when they will arrive.

Find the problem. Build the capability. Demonstrate the progress. Help others take the next step.